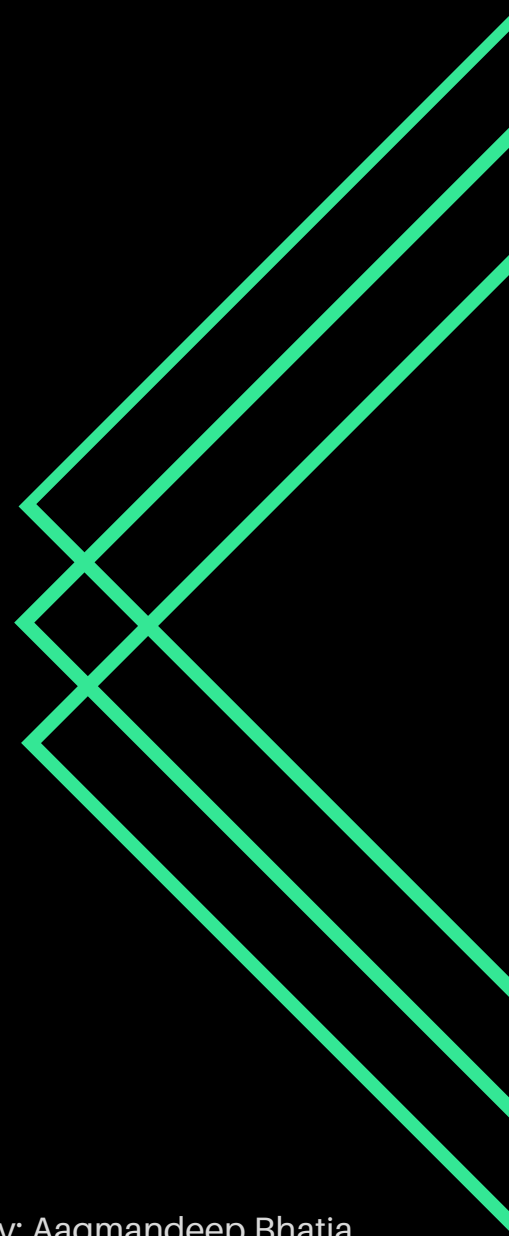




COMBINED CYCLE POWER PLANT

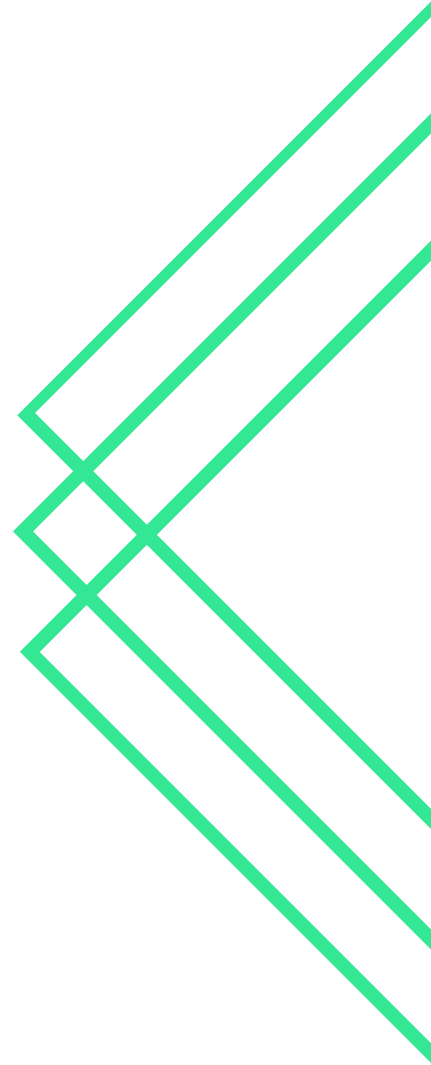
DATASET & PREDICTION



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Understanding about Business Logic Behind CCPP

The business logic behind combined cycle power plants centers on maximizing efficiency, reducing costs, and minimizing environmental impact. By leveraging both gas and steam turbines, these plants achieve higher efficiency and flexibility compared to traditional power generation methods. This translates to lower fuel consumption, enhanced grid stability, and increased profitability through electricity sales, capacity payments, and environmental incentives. Overall, combined cycle power plants offer a compelling value proposition with their superior performance, reliability, and sustainability, making them a preferred choice for electricity generation in today's energy landscape. In addition to electricity sales, combined cycle power plants earn revenue through various mechanisms such as Power Purchase Agreements (PPAs), capacity payments, and ancillary services. PPAs provide stable revenue streams through long-term contracts with utilities or other buyers, mitigating risks associated with fluctuating electricity prices. Capacity payments ensure plants are compensated for their availability during periods of peak demand, while participation in ancillary service markets allows them to provide essential grid stability services and generate additional income.



Features considered & Why?

For assessing the model's performance, all available features were utilized, along with four additional feature combinations, each comprising:

- Temperature (T) ranging from 1.81°C to 37.11°C,
- Ambient Pressure (AP) ranging from 992.89 to 1033.30 millibars,
- Relative Humidity (RH) ranging from 25.56% to 100.16%,
- Exhaust Vacuum (V) ranging from 25.36 to 81.56 cm Hg,

with the target variable being the net hourly electrical energy output (PE) spanning from 420.26 to 495.76 MW.

Additionally, a new feature named "heat_index" was engineered by amalgamating the average temperature and relative humidity.

The variables were considered due to their high correlation with the target variable PE, ensuring a comprehensive analysis of factors influencing power generation efficiency.

Model Considered

Regression Models

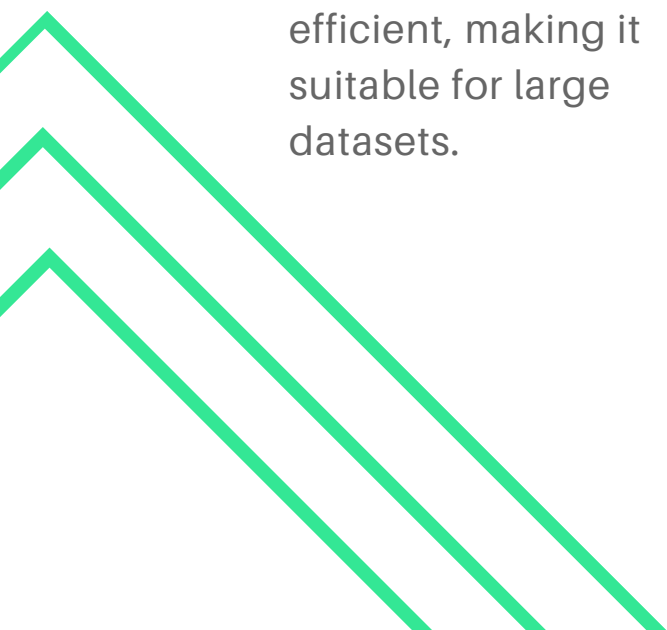
1. Linear Regression:

- **Interpretability:** Useful when we need to understand the impact of each feature on the target variable in a straightforward manner.
- **Assumption of Linearity:** When there's a linear relationship between features and target, linear regression provides a simple yet effective model.
- **Ease of Implementation:** Quick to implement and computationally efficient, making it suitable for large datasets.

1. Random Forest

Regressor:

- **Flexibility:** Can capture non-linear relationships and interactions between features and target variable effectively.
- **Robustness:** Less sensitive to outliers and noise in the data compared to linear regression, providing more reliable predictions.
- **Ensemble Approach:** Combines multiple decision trees, reducing the risk of overfitting and improving generalization performance.
- **Feature Importance:** Provides insight into feature importance, helping identify key predictors in the dataset.



Technical Approach

Technical approach (data preprocessing steps, feature engineering techniques, evaluation metrics, etc.).

Data Preprocessing:

- The dataset for this regression problem was meticulously cleaned, devoid of any null values, and largely within an acceptable range. Only a handful of instances slightly deviated from the range, with just 2-3 values falling marginally below it. Consequently, these outliers were deemed inconsequential and retained in the dataset.

Data Engineering:

- In pursuit of refining the feature set, a new attribute was introduced by amalgamating average temperature and relative humidity, resulting in the creation of a 'heat-index' column. This new feature was derived using a formulated expression to encapsulate the combined influence of temperature and humidity on the dataset.

Evaluation Metrics:

- To gauge the performance of the regression model, two key evaluation metrics were employed: Root Mean Squared Error (RMSE) and R-squared (R²) score. RMSE, indicative of the deviation between predicted and actual values, was selected as a primary metric due to its intuitive interpretation – lower values signifying better performance. Complementing RMSE, the R² score, bounded between -1 and 1, served as an additional measure of model accuracy, with higher scores indicating superior predictive capability.

References:

1. <https://www.youtube.com/watch?v=qG7SlVfAuPA>
2. <https://www.geeksforgeeks.org/heat-index-formula/>

